Chest Pain Assessment Emergency Department

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TARGET AUDIENCE and SETTING

All clinical staff working within Monash Health Emergency Departments (ED).

This guideline is applicable to all Monash Health ED patients who present with chest pain.

PURPOSE

The purpose of this procedure is to guide practice in assessment and initial management of patients presenting to the Emergency Department with Chest Pain.

STANDARD REQUIREMENTS

When undertaking any clinical interaction with a patient, staff are expected to;

- Perform routine hand hygiene. Refer to the Hand Hygiene Procedure.
- Introduce themselves to the Patient and Carer/ Family if in attendance
- Check patient identification. Refer to the Patient Identification Procedure.
- Obtain consent as per the Consent to Medical Treatment Procedure.
- Keep the patient/carer informed and involve them in decision making.
- Document interaction in the electronic medical record or health record using black pen; including date, time, signature and designation.

PROCEDURE

Initial Management

- Triage to a monitored cubicle, Australasian Triage Scale (ATS) category usually 2, but may vary depending on the nurse's perceived likelihood of acute coronary syndrome (ACS).
- Initiation of cardiac monitoring and perform a 12 lead ECG
- Performance of a full set of vital signs including pain score and level of alertness.
- Consider placement of a peripheral intravenous catheter.

Vital signs

Complete a full set of vital signs including pain score.
If these are abnormal (heart rate (HR) less than 60 or more than 130, blood pressure (BP) less than 90 systolic, oxygen saturation less than 90% on room air) call ED medical emergency team (MET).

Electrocardiogram (ECG) review:

Performance and reporting must be consistent with the requirements detailed in the PROMPT procedure: ECG performance and reporting in the Emergency Department

In particular:

- all patients triaged as 'Chest Pain – Cardiac' must have an ECG performed and reported by the area Medical Team Leader within ten minutes;
- all patients triaged as one of the following: 'chest pain – other', collapse, cardiac arrhythmia, shortness of breath, diabetes-related illness, apparent stroke or who meet ED MET criteria must have an ECG performed and reported by the area Medical Team Leader within thirty minutes.

Initial Assessment

- Complete thorough history and clinical examination (as per Clinical Examination 8th Edition Talley and O'Connor, or similar) keeping in mind flags for early life threats including myocardial infarction, pneumothorax, pulmonary embolus and aortic dissection.
- Ongoing pain requires prompt intervention from the medical team leader to initiate pain relief measures (e.g. aspirin, glyceryl trinitrate, morphine) with repeat ECGs being done ten minutely until pain is resolved
- Chest X-ray (CXR); usually diagnostic for pneumothorax; may occasionally support the diagnoses of aortic dissection or pulmonary embolism but is more useful for providing information on associated problems or conditions such as fluid overload or chest infection. Issues of timing, mobile versus not, nurse escort to X-ray versus not are case-by-case decisions.
- Obtain blood for a troponin level (gold-topped tube) along with other pathology tests as clinically indicated, most often including full blood count (FBE) and urea electrolytes and creatinine (UEC). Perform baseline coagulation studies and D-dimer if considering Pulmonary Embolus (PE) see below.
    - The need and timing of serial troponin levels is detailed in the specific PROMPT procedure: Assessment of Adult Patients for Presence of Myocardial Injury or Myocardial Infarction. A minority of patients with pulmonary embolism and aortic dissection also have abnormally elevated troponin levels.
    - Note: once a troponin test has been ordered for a patient with a clinical suspicion of acute coronary syndrome the request cannot be cancelled without discussion with the Consultant in charge ED.

- Early or subsequent presence of 'red flags' must prompt immediate senior medical involvement, in particular:
  - Ongoing chest pain despite appropriate initial treatment.
  - Presence of vital signs meeting MET call criteria.
  - Dynamic ECG changes.
  - Development of neurological symptoms or signs.

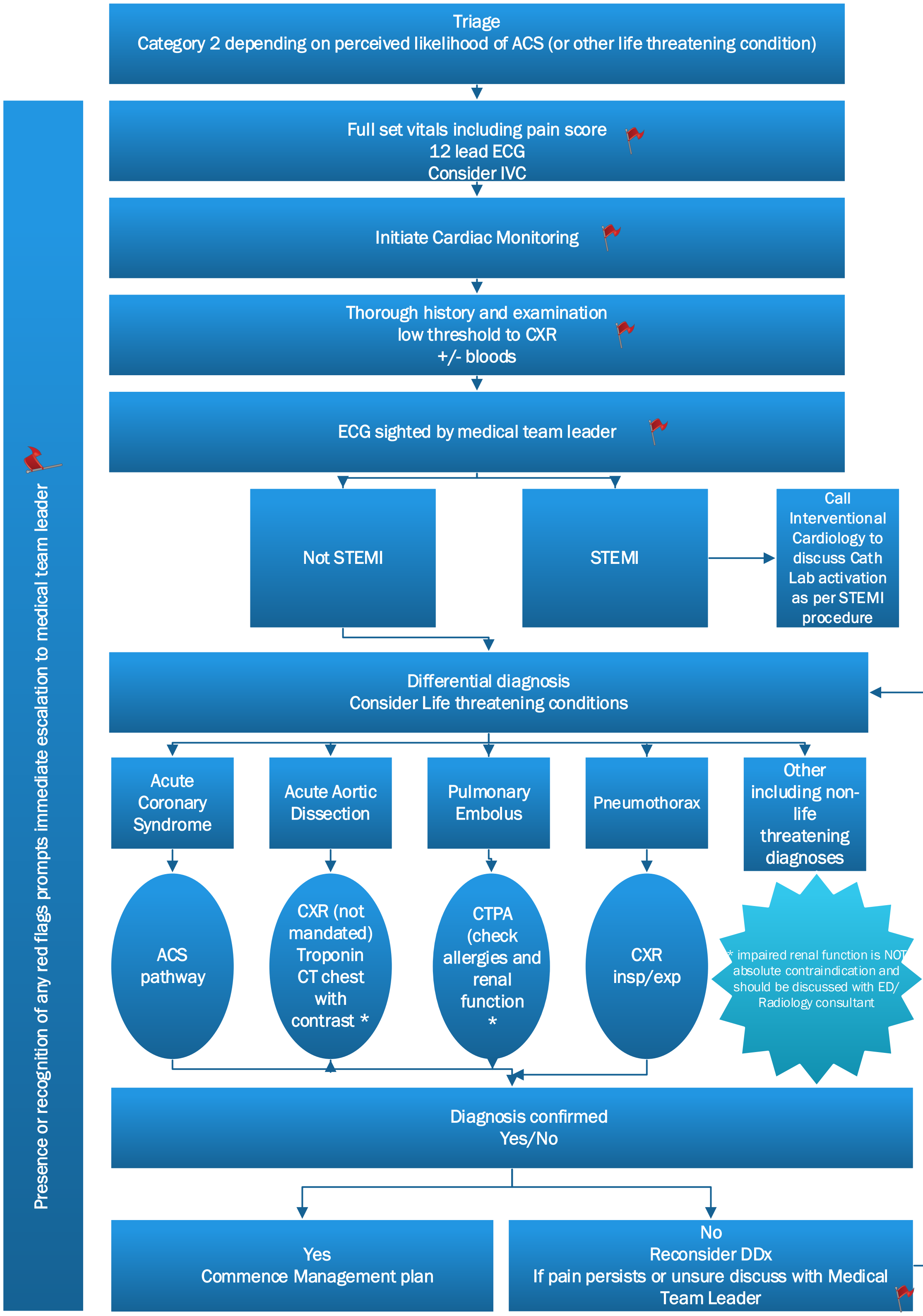
Choice of pathway/ investigations in line with working diagnosis
Of the four early life threats which most frequently present with chest pain, myocardial infarction is the most common. Pulmonary embolism, aortic dissection and pneumothorax are all relatively rare.

Non-life-threatening causes for chest pain that are to be considered include gastro-oesophageal pathology, pleuritis and other pulmonary disease, muscular and skeletal causes including costochondritis, and herpes zoster.

When considering the likely presence of any of these conditions, general risk factors and the components of published risk stratification tools must be considered. Some of these are described in detail in the PROMPT procedures or guidelines for the particular conditions.

Other considerations:
- People with potentially serious conditions such as cardiac failure, chest infection, exacerbations of chronic obstructive pulmonary disease (COPD)/ asthma and rapid tachyarrhythmia may have chest pain, but it is uncommonly the most prominent symptom (in comparison, for example, with shortness of breath or palpitations).
- The failure to make a definite diagnosis for many presentations of chest pain is well documented. This is acceptable provided reasonable measures (related to likely risk) have been taken to exclude the early life threat and other potentially serious conditions.

The following flow diagram outlines the general initial assessment and investigative approach to ED patients with chest pain.


Triage
Category 2 depending on perceived likelihood of ACS (or other life threatening condition)
Full set vitals including pain score
12 lead ECG
Consider IVC
Initiate Cardiac Monitoring
Thorough history and examination
low threshold to CXR
+/- bloods
ECG sighted by medical team leader
Not STEMI
STEMI
Call Interventional Cardiology to discuss Cath Lab activation as per STEMI procedure
Differential diagnosis
Consider Life threatening conditions
Acute Coronary Syndrome
Acute Aortic Dissection
Pulmonary Embolus
Pneumothorax
Other including non-life threatening diagnoses
ACS pathway
CXR (not mandated) Troponin CT chest with contrast *
CTPA (check allergies and renal function *
CXR insp/exp
* impaired renal function is NOT absolute contraindication and should be discussed with ED/ Radiology consultant
Diagnosis confirmed
Yes/No
Yes
Commence Management plan
No
Reconsider DDx
If pain persists or unsure discuss with Medical Team Leader
Presence or recognition of any red flags prompts immediate escalation to medical team leader

Aortic Dissection

This life threatening condition remains a challenge to diagnose.

Investigation for suspected Aortic dissection follows the procedure:

Suspected thoracic aortic dissection assessment Emergency Department

Summarised:

CXR recommended but not mandated.

ECG, troponin level.

CT scan (chest) considered first with contrast- providing no allergy or impaired renal function. NOTE: impaired renal function is NOT absolute contraindication and must be discussed with Radiology/ED Consultant to consider alternatives or renal protection using intravenous hydration with 0.9% saline as required.**

Echocardiogram appropriate in some situations.

Acute Coronary Syndrome

Follow pathway defined in procedure:

Assessment of Adult Patients for Presence of Myocardial Injury or Myocardial Infarction

Pulmonary Embolus

Investigation for suspected pulmonary embolus follows pathway defined in procedure:

Pulmonary embolism diagnosis and management

Initial Investigation Pathways- Prioritising Patient Care

Summarised:

A CT pulmonary angiogram (CTPA) is investigation of choice provided that:

- Patient is stable,
- Has no allergy to contrast,

Renal function is not impaired. NOTE: impaired renal function is NOT absolute contraindication and must be discussed with Radiology/ED Consultant to consider alternatives or renal protection using intravenous hydration with sodium chloride 0.9% as required.**

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Any female patient who is of childbearing age (<55) or with a contraindication to CTPA (renal function, contrast allergy) must be discussed with the nuclear medicine physician (in hours and out of hours) for consideration of V/Q or perfusion only scan after acquisition of a chest x-ray and BhCG. The appropriate investigation will be selected by the nuclear medicine physician.

Alternative investigations to be guided by Radiology/ED consultant/ +/- nuclear medicine/cardiology.

Pneumothorax

Investigation for suspected pneumothorax follows pathway defined in procedure:

Pneumothorax (spontaneous) Short Stay Unit/Extended Care Unit

[Blunt Chest Trauma in the elderly: Assessment and Disposition]

[Initial Investigation Pathways- Prioritising Patient Care]

CXR- Inspiratory/Expiratory

CT chest for patients >65 who have presented to ED with a primary complaint of blunt chest trauma or evidence of blunt chest trauma

**Intravascular iodinated contrast media is given to any patient regardless of renal function status if the perceived diagnostic benefit to the patient justifies this administration.

The risk of intravenous contrast media related acute kidney injury (CI-AKI) is likely to be non-existent for patients with eGFR > 45 mL/min/1.73m2. No special precautions are recommended in this group prior to or following intravenous administration of iodinated contrast media.

The risk is of intravenous CI-AKI is also very likely to be low or non-existent for patients with eGFR 30 - 45 mL/min/1.73m2. Patients with impaired function in this range that is acutely deteriorating may benefit from peri-procedural hydration.

In patients with eGFR < 30 mL/min/1.73m2 or actively deteriorating renal function (acute kidney injury) careful weighing of the risk versus the benefit of iodinated contrast media administration needs to be undertaken. Consideration shall be given to peri-procedural renal protection using intravenous hydration with sodium chloride 0.9%. However, severe renal function impairment must not be regarded as an absolute contraindication to medically indicated iodinated contrast media administration

Discharge
Discharge planning and [Clinical Assessment Prior to Discharge Home] to be completed in line with relevant procedures (linked and below)

In particular the following must be clear and included in clinical documentation:

- Diagnosis confirmed, life threatening conditions excluded and/or further investigations/follow up defined
- Pain controlled
- Plan in place/referrals made

DEFINITIONS

ED Emergency Department

ECG Electrocardiograph

STEMI S-T Elevation Myocardial Infarction

MI Myocardial Infarction

ATS Australian Triage Scale

ACS Acute Coronary Syndrome

HR Heart Rate

BP Blood pressure

ED MET Emergency Department Medical Emergency Team

ISBAR: Identify, Situation, Background, Assessment, Request

HMO Hospital Medical Officer

CXR Chest X-Ray

CT Computerised Tomography

EMR Electronic Medical Record

EDACS Ed Assessment of Chest Pain Score

CTPA Computerised Tomography Pulmonary Angiogram

AKI Acute Kidney Injury

GFR Glomerular Filtration rate

V/Q Ventilation/Perfusion

KEY STANDARDS, GUIDELINES OR LEGISLATION

National Heart Foundation of Australia and New Zealand: Australian Clinical Guideline for the Management of Acute Coronary Syndromes 2016.

Assessment Care Planning and Discharge

Acute Asthma Management (Adult)

Emergency Department Patients diverted/discharged with advice by Triage Nurse

Mental Health patient assessment, treatment, transfer and discharge in the Emergency Department

http://www.imagingpathways.health.wa.gov.au/index.php/imaging-pathways/cardiovascular/spontaneous-aortic-dissection#pathway

https://www.ranzcr.com/college/document-library/ranzcr-iodinated-contrast-guidelines

REFERENCES

Evangelista A, Isselbacher EM, Gleason TG et al. Insights From the International Registry of Acute Aortic Dissection, a 20-year Experience of Collaborative Clinical Research. *Circulation*. 2018; **137**:1846-1860.

Talley NJ, and O'Connor S, Clinical Examination 8th Edition. *Elsevier Australia*, 2017

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